

PRANJAL KARAN

Software Engineer | Full Stack Developer

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Professional Summary

Computer Science undergraduate (CGPA 8.70) and Azure-certified developer with strong fundamentals in C++, JavaScript, and data structures. Proven ability to design and ship high-performance full-stack applications—utilizing React, Node.js, and IoT integrations—while optimizing database latency and collaborating seamlessly across teams to deliver scalable problem-solving solutions.

Education

VIT Bhopal University – Bachelor of Technology, Computer Science & Engineering — Sehore, MP Sept 2023 – May 2027 CGPA: 8.70/10

Technical Skills

Languages: C++, JavaScript (ES6+), SQL

Frontend: React, HTML5, CSS3, Tailwind CSS v4, Vite, Responsive Web Design

Backend & APIs: Node.js, Express.js, RESTful API Design, WebSockets, JWT Authentication, Clerk Auth

Databases: MongoDB (Mongoose, Aggregation Pipelines), MySQL, PostgreSQL

AI Tools: Claude, ChatGPT, GitHub Copilot, Antigravity

Tools: VS Code, ESP8266, Blynk IoT, Git, GitHub, Vercel, SDLC

Soft Skills: Agile Problem-Solving, Cross-Functional Collaboration, Adaptability, Team Leadership

Projects

VibeChat – Real-Time Full-Stack Messaging Platform — GitHub 2026 React 19, Tailwind v4, Zustand, Node.js, MongoDB, Socket.io, Clerk, ImageKit, Docker

- **Engineered** a real-time messaging application achieving **sub-100ms** message distribution and interactive live presence indicators utilizing custom Socket.io server channels.
- **Accelerated** database query performance by replacing N+1 queries with a **6-stage Mongoose aggregation pipeline** and \$lookup joins, eliminating redundant roundtrips.
- **Built and tested** a lightweight **1.17MB** production bundle via Vite 8 featuring Clerk OAuth session gates, ImageKit lazy-loading pipelines, and a developer Mock Auth fallback.

HealConnect – Full-Stack Healthcare Monitoring Platform — GitHub Jan 2026 – Present Next.js, React, C++/Arduino, Firebase, Firestore, WebSockets, Blynk IoT, Gemini

- **Architected** a **13.1K-line** Next.js platform spanning **44 pages** and **23 reusable components**, integrating Firebase phone auth, EmailJS, and Gemini AI APIs.
- **Designed** C++/Arduino firmware for wearable ECG kits, achieving sub-second live data streaming via Blynk IoT, WebSockets, and Firestore onSnapshot listeners.
- **Delivered** role-based dashboards (Patient, Doctor, Admin) backed by **3 secure serverless APIs** and async polling loops for automated prescription processing.

LiveClass – Real-Time Video Learning Platform — GitHub 2026 MongoDB, Express.js, React, Node.js, JWT, REST APIs

- **Developed** a scalable MERN application managing the complete lifecycle of live video classroom sessions with persistent historical data logging in MongoDB.
- **Secured 100%** of authenticated REST API endpoints by implementing JWT authentication and Bcrypt password hashing across backend services.

Achievements & Certifications

Achievements: National Qualifier – Cleared Round 1 of TCS CodeVita, a national-level competitive coding challenge

Certifications: MERN Full Stack Certification (Ethnus & Codemithra) – [ID: RMGJBVLS] — Microsoft Certified: Azure Data Fundamentals (Score: 955/1000) — ServiceNow Virtual Internship (SmartBridge / AICTE) – Configured workflows & validated IT operations using Agentic AI and ATF — AEO Fundamentals (HubSpot)

Leadership: Co-Lead (Social Media), Linpack Club, VIT Bhopal – **Increased** event engagement by **30%** through targeted digital campaigns. Received a Letter of Appreciation for outstanding teamwork.